PackLite: Compressed Model Checkpointing

Abstract

PackLite compresses neural network checkpoints using delta encoding between consecutive training steps, cutting checkpoint storage by sixty percent without any loss in model accuracy.

Introduction

Large model checkpoints consume enormous storage over long training runs. PackLite reduces this footprint by exploiting redundancy between consecutive checkpoints rather than storing each one in full.

Method

PackLite stores only the difference between each checkpoint and the previous one, using sparse delta encoding to keep the diffs small.

Results

Across five large language model training runs, PackLite reduced total checkpoint storage from 2.1 terabytes to approximately 840 gigabytes, with no measurable change in downstream task accuracy.